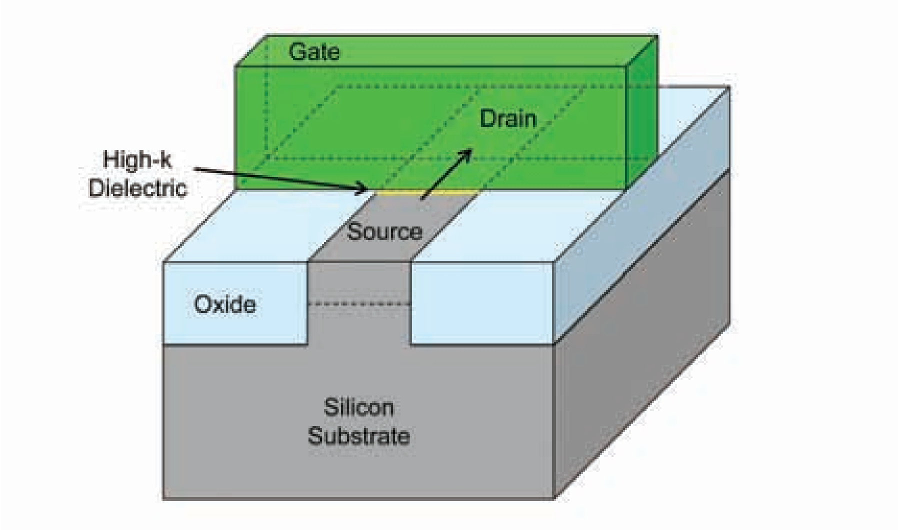
transistor has evolved and become smaller and smaller as integrated circuits required drastic size reductions. This is necessary for many reasons – more power is being demanded in increasingly smaller packages. 8 years ago, cellphones had very basic processors that were a couple of cuts above embedded processors, since their functionality was very limited. Today, cellphones are processing powerhouses, with dual core processors running at previously un-thought of speeds. More performance means more transistors, and the package also needs to be smaller creating a myriad of problems – thermal dissipation, current leakage, and even manufacturing problems. To those who don't know, think of a transistor as a very fast electrical switch with three terminals. As with a circuit, when it's switched on, current flows signifying a 1 in binary, and when switched off, current ceases to flow, signifying a 0.

Therefore in off position, the transistor's source is supplied with a charge. This source is a conductive portion of a silicon wafer. However, the channel that is a link between the source and drain acts as an insulator preventing electrons from flowing through it and the computer reads the result as a 0. For the transistor to turn on, a charge is applied to the gate that pulls electrons towards it and on to the channel that becomes conductive as a result, creating a clear electrical circuit between source and drain. This is read by a computer as a 1. One of the problems as you go on shrinking the design is as components that are supposed to conduct and insulate in turn become smaller, they tend to become less effective as insulation, as current leakage becomes more of an issue. This current leakage means that there is a current flow even when the circuit is in an open (off) state. This leads to a much smaller gap between voltages in on and off state and this can lead to errors in the system detecting between an on and an off. For Intel, the gate insulator became so thin that there were problems with current leakage, and this was around the time when we saw the infamous Pentium 4 processors that were renowned as miniature heaters.

Traditional Planar Transistor



Traditional 2-D planar transistors form a conducting channel in the silicon region under the gate electrode when in the "on" state

Intel managed to partially mitigate this by continuously using new and improved materials – strained silicon, a hafnium-based insulator, high-K metal gates – chances are you've heard of at least some of these exotic sounding handles, but all used for very similar purposes – enabling better and more precise control on the on and off states, and prevention of current leakages and voltage retention within the silicon substrate itself. Therefore, there are physical limits and material limits to how small transistors can get, and ultimately how powerful the resultant processor can get. With current leakage comes inefficiency and heating, and these only compound the problem.

Rationally, there are two ways to solve this problem. One is to obviously let through more electrons, while the second is to reduce the leakage current. To get more electrons through, one needs to make the channel more conductive and therefore increase the volts supplied to it. However, power consumption also increases. The second method is to increase the surface area of the conductive material – because a larger charged surface area would attract more electrons and therefore transmit more current. And this is exactly what Intel did with the Tri-gate transistor – made